```python
import pandas as pd
data=pd.read_csv("student-lifestyle-and-stress-dataset.csv")
print(data.head())
print(f"Number of duplicate rows: {data.duplicated().sum()}")
data=data.drop_duplicates()
print(f"Number of duplicate rows after removal: {data.duplicated().sum()}")
print(data.isnull().mean()*100)
data['Sleep_Hours'] = data['Sleep_Hours'].fillna(data['Sleep_Hours'].median())
data['Study_Hours'] = data['Study_Hours'].fillna(data['Study_Hours'].median())
data['Social_Media_Hours'] = data['Social_Media_Hours'].fillna(data['Social_Media_Hours'].median())
data['Attendance'] = data['Attendance'].fillna(data['Attendance'].median())
data['Exam_Pressure'] = data['Exam_Pressure'].fillna(data['Exam_Pressure'].median())
data['Month'] = data['Month'].fillna(data['Month'].median())
data['Family_Support'] = data['Family_Support'].fillna(data['Family_Support'].median())
data['Student_Type'] = data['Student_Type'].fillna(data['Student_Type'].mode()[0])
print(data.isnull().sum())

from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
data['Student_Type']=label_encoder.fit_transform(data['Student_Type'])
x=data.drop(columns=["Stress_Level"])
y=(data[['Stress_Level']])

from sklearn.preprocessing import StandardScaler
x_scaler=StandardScaler()
x_scaled=x_scaler.fit_transform(x)

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x_scaled,y,test_size=0.2,random_state=42)

from sklearn.linear_model import LogisticRegression
model=LogisticRegression()
model.fit(x_train,y_train)
y_predict=model.predict(x_test)
print('coefficient :',model.coef_)
print('intercept:',model.intercept_)

from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test,y_predict)
print('accuracy :',accuracy)

import joblib
joblib.dump(model, "model.pkl")
joblib.dump(x_scaler, 'x_scaler.pkl')
joblib.dump(label_encoder, "label_encoder.pkl")
```

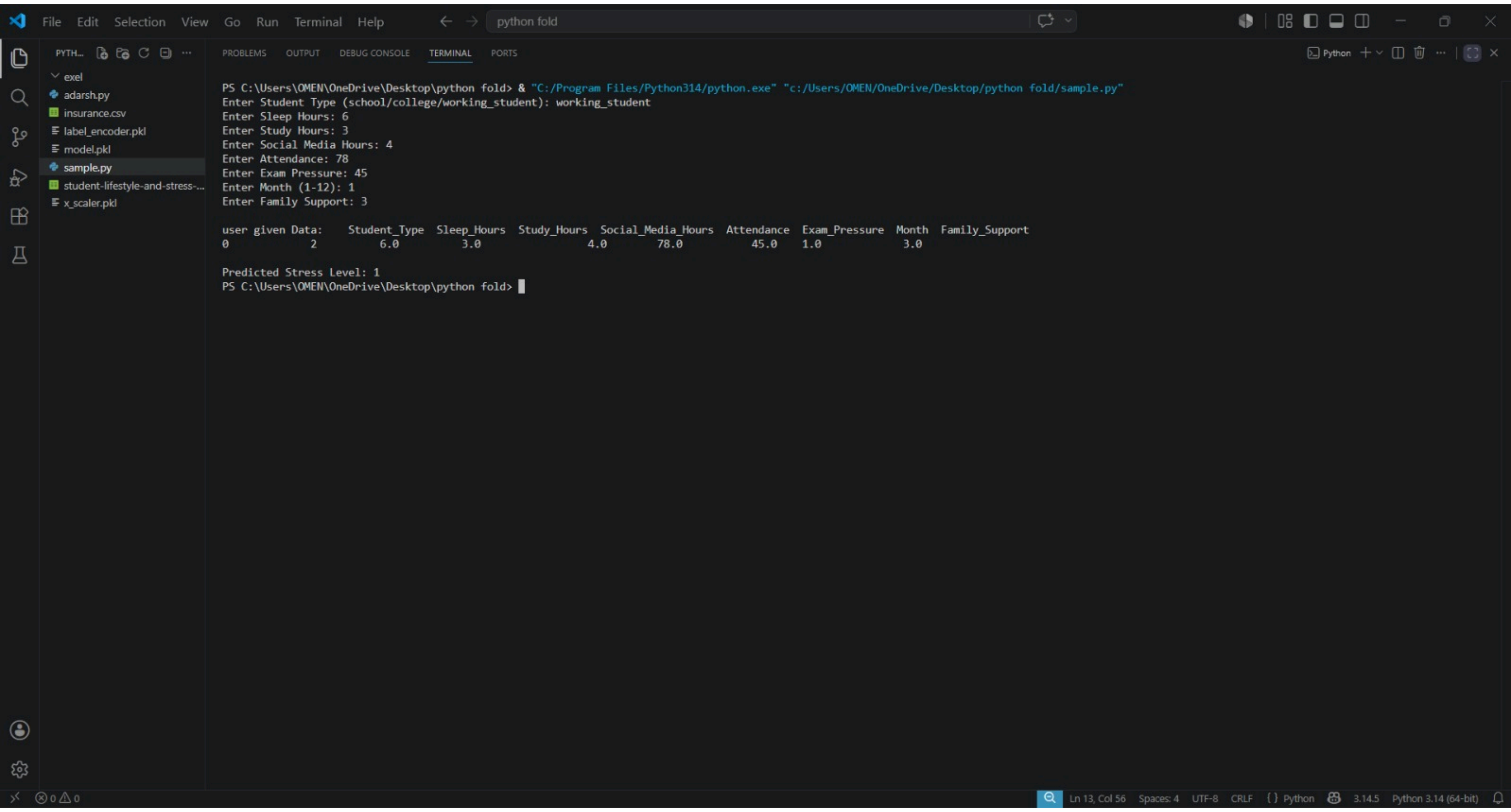
```
PS C:\Users\OMEN\OneDrive\Desktop\python fold> & "C:/Program Files/Python314/python.exe" "c:/Users/OMEN/OneDrive/Desktop/python fold/adarsh.py"
  Student_Type  Sleep_Hours  Study_Hours  Social_Media_Hours  Attendance  Exam_Pressure  Month  Family_Support  Stress_Level
0       school     6.868702     1.711722            3.176942         NaN            8.0    7.0             2.0             1
1       school     8.519088     3.251084            3.880787   93.978465            6.0    4.0             3.0             1
2      college     4.498770     6.306885            2.936172   64.421253            7.0    1.0            12.0             1
3       school     8.591223     2.384922            5.222832   81.868960            2.0    7.0             7.0             0
4      college     5.329293     9.345179            7.815869   85.847982            5.0    6.0            10.0             1
Number of duplicate rows: 24
Number of duplicate rows after removal: 0
Student_Type          4.910504
Sleep_Hours           5.232376
Study_Hours           4.918355
Social_Media_Hours    5.142095
Attendance            5.118543
Exam_Pressure         4.985084
Month                 5.063589
Family_Support        5.157796
Stress_Level          0.000000
dtype: float64
Student_Type          0
Sleep_Hours           0
Study_Hours           0
Social_Media_Hours    0
Attendance            0
Exam_Pressure         0
Month                 0
Family_Support        0
Stress_Level          0
dtype: int64
C:\Users\OMEN\AppData\Roaming\Python\Python314\site-packages\sklearn\utils\validation.py:1352: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape o
f y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
coefficient : [[-0.02050641 -0.33631146  0.63451352  0.31305693 -0.10475569  1.65836813
  -0.36624527 -0.19763972]]
intercept: [-1.36219389]
accuracy : 0.8137755102040817
PS C:\Users\OMEN\OneDrive\Desktop\python fold>
```

```python
import joblib
model = joblib.load("model.pkl")
x_scaler = joblib.load("x_scaler.pkl")
encoder = joblib.load("label_encoder.pkl")

student_type = input("Enter Student Type (school/college/working_student): ")
sleep_hours = float(input("Enter Sleep Hours: "))
study_hours = float(input("Enter Study Hours: "))
social_media_hours = float(input("Enter Social Media Hours: "))
attendance = float(input("Enter Attendance: "))
exam_pressure = float(input("Enter Exam Pressure: "))
month = float(input("Enter Month (1-12): "))
family_support = float(input("Enter Family Support: "))
student_type_encoded = encoder.transform([student_type])[0]

import pandas as pd
new_data = pd.DataFrame([[student_type_encoded,sleep_hours,study_hours,social_media_hours,attendance,exam_pressure,month,family_support]], columns=['Student_Type','Sleep_Hours','Stu
print("\nuser given Data:",new_data)
new_data_scaled = x_scaler.transform(new_data)
predicted_stress = model.predict(new_data_scaled)

print("\nPredicted Stress Level:", predicted_stress[0])
```



File Edit Selection View Go Run Terminal Help
python fold
PYTH...
exel
adarsh.py
insurance.csv
label_encoder.pkl
model.pkl
sample.py
student-lifestyle-and-stress-...
x_scaler.pkl
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\OMEN\OneDrive\Desktop\python fold> & "C:/Program Files/Python314/python.exe" "c:/Users/OMEN/OneDrive/Desktop/python fold/sample.py"
Enter Student Type (school/college/working_student): working_student
Enter Sleep Hours: 6
Enter Study Hours: 3
Enter Social Media Hours: 4
Enter Attendance: 78
Enter Exam Pressure: 45
Enter Month (1-12): 1
Enter Family Support: 3
user given Data: Student_Type Sleep_Hours Study_Hours Social_Media_Hours Attendance Exam_Pressure Month Family_Support
0 2 6.0 3.0 4.0 78.0 45.0 1.0 3.0
Predicted Stress Level: 1
PS C:\Users\OMEN\OneDrive\Desktop\python fold>
Ln 13, Col 56 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

Sorry for using more than 100 dataset
this help me to use data.duplicate
data.is null () etc.

Problem definition.

Problem: Students often face stress due to lifestyle factors such as sleep, study habits, social media usage, attendance, exam pressure and family support.
and this is a classification problem.

Data Description

Data set: is from kaggle. I will share the link.
student-lifestyle and stress-dataset.csv.

Features (inputs):

- sleep - Hours: numerical.
- student - Type (school/college/working_student) → categorical
- study - Hours → numerical
- social - Media - Hours → numerical
- Attendance → numerical.
- exam - Pressure → numerical.
- Month → numerical.
- Family - Support → numerical.

Target variable: stress_level 0 → No
1 → Yes

Data Preprocessing

- Handle missing values:
  - For numerical → Fillna (data[ ].median())
  - For categorical → Fillna (data[ ].mode())

Remove duplicates: data = data.drop_duplicates().

Label Encoding → student_Type.

Feature scaling: applied to x become x-scaled
not to y (classification) → Target variable

Train-test split:
80% training, 20% testing using train-test-split.

Model Building

- Logistic Regression.
from sklearn.linear_model import LogisticRegression.

Justification.
Because this is a classification dataset.

Model Evaluation
from sklearn.metrics importing mean_squared_error.
we can predict error
and importing numpy we can use square root to predict rmse

Interpretation:
~~MSE measures average squared difference between predicted and actual stress levels.~~
~~RMSE expresses this error in the square root.~~

From dataset:

~~MSE:~~	~~0.18622~~
~~RMSE:~~	~~0.43153~~

After traing the logistic Regression model, the evaluation was done using accuracy.
from sklearn.metrics import accuracy_score
accuracy: 0.8137

Conclusion:
Outcomes: logistic Regression successfully modeled stress levels based on lifestyle features.

- Limitation:
Data set size and missing values may reduce reliability

- Future improvements:
  - Collect larger datasets.
  - Try to use different models we have studied.

Reflections.

This project helped me understand how machine learning can be used to study real-life problems like student stress. I learned how important it is to prepare the dataset before building a model. Handling missing values, removing duplicates, encoding categorical data, and scaling features showed me that preprocessing is a key step for getting good results. I also realized that even small mistakes in preprocessing can affect the accuracy of the model.

One challenge I faced was choosing the right evaluation method. At first, I used mean squared error and root mean squared error, which are more common in regression problems. Later, I understood that for classification tasks, metrics like accuracy are more suitable. This taught me to think carefully about which performance measures fit the problem.

I also understand ~~one more~~ difference between Linear Regression and Logistic Regression.

The main concepts I learned were data cleaning, train-test splitting, model training, and evaluation. Building a logistic regression model gave me confidence in applying machine learning to categorical outcomes. Overall, this project gave me a clear picture of the full machine learning workflow and confidence to deal different datas.